



Asthma

Ethiopia

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Identity

Model	Asthma
Country	Ethiopia
Reference parameter set	reference-parameter-set:ncd-asthma:synthetic-v1
Scenario registry	scenario-registry:ncd-asthma:reference-synthetic-v1
Source model hash	94c67579de128117

Reference Case

Reference case	Ethiopia asthma inputs
Country code	ETH
Output years	2000-2100
Engine runtime	2000-2101
Data bundle	05f1380ee9926dff

Provenance

Record	Source	Citation
Age-specific fertility rate	World Population Prospects; 2022 revision, medium variant; United Nations Department of Economic and Social Affairs, Population Division	United Nations, Department of Economic and Social Affairs, Population Division. World Population Prospects 2022.
Net migration rate	World Population Prospects; 2022 revision, general demographic indicators; United Nations Department of Economic and Social Affairs, Population Division	United Nations, Department of Economic and Social Affairs, Population Division. World Population Prospects 2022.
Life table mortality by age and sex	World Population Prospects; 2022 revision, medium variant; United Nations Department of Economic and Social Affairs, Population Division	United Nations, Department of Economic and Social Affairs, Population Division. World Population Prospects 2022.
Population by single year of age and sex	World Population Prospects; 2022 revision, medium variant; United Nations Department of Economic and Social Affairs, Population Division	United Nations, Department of Economic and Social Affairs, Population Division. World Population Prospects 2022.
Sex ratio at birth	World Population Prospects; 2024 revision, medium variant; United Nations Department of Economic and Social Affairs, Population Division	United Nations, Department of Economic and Social Affairs, Population Division. World Population Prospects 2024.
Healthy population disability weights by GBD region, sex and age bin	Spectrum/GBD disability-weight defaults; Spectrum GBDModData disability-weight CSV; Avenir Health; Institute for Health Metrics and Evaluation; World Health Organization	Avenir Health. Spectrum Manual; GBD disability-weight defaults as bundled in Spectrum GBDModData.
Asthma high-dose beclometasone disability impact	Spectrum/OneHealth asthma disease association file; Spectrum 5 Asthma workbook; Avenir Health; World Health Organization	Avenir Health. OneHealth Manual. NCD module section; Avenir/WHO NCD OneHealth guide.
Asthma high-dose beclometasone population in need	Spectrum/OneHealth asthma disease association file; Spectrum 5 Asthma workbook; Avenir Health; World Health Organization	Avenir Health. OneHealth Manual. NCD module section; Avenir/WHO NCD OneHealth guide.
Asthma inhaled short-acting beta agonist disability impact	Spectrum/OneHealth asthma disease association file; Spectrum 5 Asthma workbook; Avenir Health; World Health Organization	Avenir Health. OneHealth Manual. NCD module section; Avenir/WHO NCD OneHealth guide.
Asthma inhaled short-acting beta agonist population in need	Spectrum/OneHealth asthma disease association file; Spectrum 5 Asthma workbook; Avenir Health; World Health Organization	Avenir Health. OneHealth Manual. NCD module section; Avenir/WHO NCD OneHealth guide.

Record	Source	Citation
Asthma low-dose beclometasone disability impact	Spectrum/OneHealth asthma disease association file; Spectrum 5 Asthma workbook; Avenir Health; World Health Organization	Avenir Health. OneHealth Manual. NCD module section; Avenir/WHO NCD OneHealth guide.
Asthma low-dose beclometasone population in need	Spectrum/OneHealth asthma disease association file; Spectrum 5 Asthma workbook; Avenir Health; World Health Organization	Avenir Health. OneHealth Manual. NCD module section; Avenir/WHO NCD OneHealth guide.
Asthma oral prednisolone disability impact	Spectrum/OneHealth asthma disease association file; Spectrum 5 Asthma workbook; Avenir Health; World Health Organization	Avenir Health. OneHealth Manual. NCD module section; Avenir/WHO NCD OneHealth guide.
Asthma oral prednisolone population in need	Spectrum/OneHealth asthma disease association file; Spectrum 5 Asthma workbook; Avenir Health; World Health Organization	Avenir Health. OneHealth Manual. NCD module section; Avenir/WHO NCD OneHealth guide.
NCD disease observations and intervention defaults	Spectrum/OneHealth NCD disease association files; Spectrum 5 ModData NC disease workbook set; Avenir Health; World Health Organization	Avenir Health. OneHealth Manual. NCD module section.
GBD epidemiology for NCD models	Global Burden of Disease via Spectrum/OneHealth NCD defaults; GBD/Spectrum legacy default dataset; Institute for Health Metrics and Evaluation; Avenir Health; World Health Organization	Avenir/WHO. A user guide to the NCD cost and impact modules of the OneHealth Tool. January 2018.
Risk-factor relative risks	Spectrum/OneHealth NCD relative-risk defaults; Spectrum 5 ModData NC workbook set; Avenir Health; World Health Organization	Avenir/WHO. A user guide to the NCD cost and impact modules of the OneHealth Tool. January 2018.
Risk-factor prevalence defaults	Spectrum/OneHealth NCD risk-factor defaults; Spectrum 5 ModData NC workbook set; Avenir Health; World Health Organization	Avenir Health. OneHealth Manual; Avenir/WHO. A user guide to the NCD cost and impact modules of the OneHealth Tool.
Alcohol and tobacco tax defaults	Spectrum/OneHealth NCD tax defaults; Spectrum 5 ModData NC workbook set; Avenir Health; World Health Organization	WHO. Technical briefing for Appendix 3 of the Global Action Plan for Non-Communicable Diseases: Tobacco control interventions. 28 November 2022.

Record	Source	Citation
Tobacco intervention effect-size and tax assumptions	WHO Appendix 3 tobacco technical assumptions via Spectrum/OneHealth; WHO technical brief 2022; Spectrum RiskFactorData workbook; World Health Organization; Avenir Health	WHO. Technical briefing for Appendix 3 of the Global Action Plan for Non-Communicable Diseases: Tobacco control interventions. 28 November 2022.

Provenance Gaps

Fetcher	Method	Scope
fhdb	ilo_labour_force_participation_rate	ETH; 2000
fhdb	wb_gdp_per_capita	ETH; 2000
fhdb	yll_weights	2000-2101

Internal Validity

Country reference run	pass
Result rows	66803
Data bundle	unified_api
Bundle inputs	1331
Headline checks	pass
Result hash	9cb0e3f54b273e2e

Headline Outputs

Year	Asthma prevalence state	Asthma incidence flow	Asthma mortality flow
2000	2,337,954	460,101	2,125
2001	2,620,741	469,841	2,306
2002	2,891,942	479,766	2,480
2003	3,152,965	490,214	2,645
2004	3,403,381	500,777	2,804
2005	3,644,189	512,039	2,962
2006	3,879,023	522,701	3,159
2007	4,109,418	533,506	3,365
2008	4,333,371	544,812	3,583
2009	4,554,178	555,899	3,810
2010	4,773,374	567,307	4,049
2011	4,990,331	578,382	4,290
2012	5,203,196	589,481	4,546
2013	5,409,557	600,049	4,813
2014	5,612,176	610,122	5,092
2015	5,809,297	621,026	5,382
2016	6,007,217	632,042	5,683
2017	6,200,881	643,990	6,004
2018	6,390,374	656,223	6,336
2019	6,579,002	669,166	6,677
2020	6,770,558	683,429	7,029
2021	6,960,430	697,805	7,358
2022	7,152,560	711,894	7,675
2023	7,348,336	725,975	8,005
2024	7,549,430	740,215	8,360
2025	7,751,736	754,230	8,745
2026	7,955,301	767,864	9,157
2027	8,160,326	781,229	9,582
2028	8,364,754	794,566	10,023
2029	8,571,459	807,629	10,475
2030	8,778,020	820,456	10,947
2031	8,984,502	832,778	11,429
2032	9,190,983	844,521	11,927
2033	9,393,879	856,201	12,436
2034	9,597,211	867,014	12,954
2035	9,800,104	877,782	13,493
2036	10,000,920	888,231	14,060
2037	10,199,278	898,410	14,649
2038	10,396,376	908,273	15,263
2039	10,592,344	918,036	15,904
2040	10,785,998	927,861	16,570
2041	10,976,656	937,664	17,260
2042	11,166,179	947,515	17,968
2043	11,355,146	957,173	18,687
2044	11,542,692	967,363	19,423
2045	11,728,086	977,241	20,182

Year	Asthma prevalence state	Asthma incidence flow	Asthma mortality flow
2046	11,913,981	987,226	20,978
2047	12,100,083	997,000	21,814
2048	12,284,094	1,007,105	22,688
2049	12,466,343	1,016,907	23,583
2050	12,647,438	1,026,858	24,492
2051	12,831,659	1,036,616	25,436
2052	13,011,998	1,046,591	26,409
2053	13,190,678	1,056,384	27,382
2054	13,368,945	1,065,893	28,366
2055	13,544,445	1,075,145	29,388
2056	13,723,669	1,083,914	30,463
2057	13,896,463	1,092,678	31,605
2058	14,065,209	1,100,902	32,782
2059	14,238,105	1,108,729	33,979
2060	14,407,244	1,116,200	35,199
2061	14,575,004	1,123,204	36,465
2062	14,736,280	1,130,438	37,767
2063	14,894,747	1,137,220	39,061
2064	15,049,527	1,143,490	40,368
2065	15,199,936	1,149,581	41,714
2066	15,346,230	1,155,398	43,108
2067	15,493,565	1,161,142	44,554
2068	15,633,908	1,166,779	46,055
2069	15,771,041	1,171,845	47,559
2070	15,904,065	1,177,210	49,034
2071	16,035,190	1,182,644	50,553
2072	16,164,679	1,187,975	52,103
2073	16,288,116	1,192,971	53,651
2074	16,412,740	1,197,453	55,191
2075	16,534,410	1,202,166	56,757
2076	16,652,581	1,207,030	58,335
2077	16,767,995	1,211,728	59,935
2078	16,879,638	1,216,264	61,548
2079	16,984,820	1,220,843	63,157
2080	17,088,743	1,225,094	64,689
2081	17,191,936	1,229,169	66,261
2082	17,294,975	1,233,268	67,845
2083	17,392,571	1,237,621	69,428
2084	17,488,521	1,241,676	70,975
2085	17,584,442	1,245,474	72,528
2086	17,679,939	1,249,163	74,110
2087	17,768,743	1,252,941	75,716
2088	17,851,074	1,256,045	77,315
2089	17,932,349	1,258,800	78,900
2090	18,016,377	1,261,445	80,503
2091	18,095,573	1,264,241	82,127
2092	18,169,486	1,266,725	83,742
2093	18,243,724	1,269,264	85,323

Year	Asthma prevalence state	Asthma incidence flow	Asthma mortality flow
2094	18,317,856	1,271,642	86,894
2095	18,385,502	1,273,978	88,473
2096	18,457,575	1,275,769	90,058
2097	18,524,844	1,277,695	91,696
2098	18,585,625	1,279,403	93,333
2099	18,641,604	1,280,810	94,977
2100	18,700,302	1,281,903	96,647